

Roll No: 2303A52385

SQL JOIN Practice – Tables and 10 Questions

Use the following tables for all questions.

1. departments

dept_id	dept_name	location
10	Engineering	Chennai
20	Sales	Hyderabad
30	HR	Bengaluru
40	Research	Pune
50	Finance	Mumbai

2. employees

emp_id	emp_name	dept_id	manager_id	salary	hire_date
101	Arjun	10	NULL	90000	2021-01-15
102	Priya	10	101	65000	2022-03-10
103	Karthik	20	106	55000	2022-07-20
104	Meena	30	NULL	60000	2020-11-05
105	Rahul	NULL	101	50000	2023-02-12
106	Divya	20	NULL	85000	2019-06-25
107	Sanjay	10	101	72000	2021-09-17
108	Anitha	40	NULL	78000	2020-04-30
109	Vikram	20	106	48000	2024-01-08

3. projects

project_id	project_name	dept_id	budget
201	Banking App	10	500000
202	Sales Portal	20	300000
203	AI Research	40	800000

204	Employee System	30	250000
205	Cloud Migration	10	600000
206	Audit Automation	50	350000

4. employee_project

emp_id	project_id	hours
101	201	120
102	201	100
102	205	80
103	202	110
106	202	70
107	201	90
107	205	130
108	203	150
104	204	60

Questions

1. Write an `INNER JOIN` query to display every employee's ID, name, department name and department location. Employees without a department should not appear.
2. Write a `LEFT JOIN` query to display all employees along with their department names. Employees who are not assigned to a department must also be displayed.
3. Write a `LEFT JOIN` query to find departments that do not currently have any employees.
4. Write a query using multiple joins to display the employee name, project name, department name and number of hours worked.
5. Write a query to display all projects and the employees assigned to each project. Projects without assigned employees must also appear.
6. Using a self-join on the `employees` table, display each employee's name and their manager's name. Employees without managers must also appear.
7. Write a query to find employees who are working on more than one project. Display the employee ID, employee name and total number of projects.
8. Write a query to display each department's name, number of employees and average employee salary. Departments without employees must show an employee count of 0.
9. Write a query to find employees who work on projects belonging to a department different from their own department. Display the employee name, employee department, project name and project department.
10. Write a query to find projects that have no employees assigned. Display the project ID, project name, department name and budget.

- 1) `select e.emp_id,e.emp_name,d.dept_name,d.location from employees e inner join departments d ON e.dept_id=d.dept_id;`
- 2) `select e.emp_id,e.emp_name,d.dept_name from employees e left join departments d ON e.dept_id=d.dept_id;`
- 3) `select d.dept_id,d.dept_name from departments d left join employees e ON d.dept_id=e.dept_id where e.emp_id IS NULL;`
- 4) `select e.emp_name,p.project_name,d.dept_name,ep.hours from employees e join employee_project ep ON e.emp_id=ep.emp_id join projects p ON ep.project_id=p.project_id JOIN departments d ON p.dept_id=d.dept_id;`
- 5) `select p.project_id,p.project_name,e.emp_name from projects p left join employee_project ep ON p.project_id=ep.project_id left join employees e ON ep.emp_id=e.emp_id;`
- 6) `select e.emp_name,coalesce(m.emp_name,'No manager') from employees e left join employees m ON e.manager_id=m.emp_id;`
- 7) `select e.emp_id,e.emp_name,count(ep.project_id) as totalprojects from employees e join employee_project ep ON e.emp_id=ep.emp_id GROUP by e.emp_id,e.emp_name HAVING Count(ep.project_id)>1;`
- 8) `select d.dept_name,count(e.emp_id),coalesce(avg(e.salary),0) from departments d left join employees e ON d.dept_id=e.dept_id group by d.dept_id,d.dept_name;`

```
10)select p.project_id,p.project_name,d.dept_name,p.budget from projects p join  
departments d ON p.dept_id=d.dept_id left join employee_project ep ON  
p.project_id=ep.project_id where ep.emp_id is null;
```